

Day 26: Print Dictionary Using `items()`

1. Day Number

Day 26

2. Topic Name

Dictionary Loop

3. Connection

Earlier, you created profile dictionaries containing information such as name, phone number, and city.

Today, you will learn how to print **all profile fields** using a loop instead of accessing every dictionary key separately.

4. Important Topics

- Dictionary `.items()`
 - Key
 - Value
 - for loop
-

5. Foundational Notes

A dictionary stores information as **key-value pairs**.

```
profile = {  
    "name": "Rahul",  
    "city": "Bangalore"  
}
```

Here:

- "name" is a key.
- "Rahul" is its value.
- "city" is another key.
- "Bangalore" is its value.

You can access one value using its key:

```
print(profile["name"])
```

However, when you want to process **every key and value**, you can use the dictionary method:

```
dictionary_name.items()
```

The `.items()` method provides both the key and its corresponding value during each loop iteration.

General structure:

```
for key, value in dictionary_name.items():  
    print(key, value)
```

During each loop:

1. key receives the current dictionary key.
2. value receives the value connected to that key.
3. The loop continues until all dictionary entries are processed.

The variable names do not have to be key and value, but these names make the code easier to understand.

6. Easy Example

```
student = {  
    "name": "Anita",  
    "course": "Python"  
}  
  
for field, information in student.items():  
    print(field, information)
```

Output

```
name Anita  
course Python
```

The loop runs twice because the dictionary contains two key-value pairs.

7. Problem Statement

Create a profile dictionary containing fields such as:

- Name
- Phone
- City

Use `.items()` with a for loop to print each key and its corresponding value.

Print each field in a readable format such as:

```
name: Rahul  
phone: 9876543210  
city: Bangalore
```

8. Concepts Used

Dictionary

Stores related information as key-value pairs.

.items()

Returns each dictionary key together with its value.

for loop

Repeats an operation for every entry in the dictionary.

Key and value unpacking

The two values returned during each iteration are stored in two variables:

```
for key, value in ...
```

Function

A function can receive the profile dictionary and handle the printing logic.

9. Thought Process

Think about the problem in these steps:

1. Create a dictionary for the profile.
2. Store profile fields as keys.
3. Store profile details as values.
4. Create a function that receives the dictionary.
5. Check whether the dictionary is empty.
6. Use `.items()` to obtain every key and value.
7. Print each pair in a readable format.
8. Call the function and pass the profile dictionary.

You do not need to write separate statements like:

```
print(profile["name"])  
print(profile["phone"])  
print(profile["city"])
```

A loop makes the program shorter and works even when more fields are added later.

10. Pseudocode

```
START

CREATE a function that accepts a profile dictionary

    IF the profile dictionary is empty
        PRINT "Profile is empty"
        STOP the function

    FOR each key and value in profile.items()
        PRINT key and value

CREATE a profile dictionary
    STORE name
    STORE phone
    STORE city

CALL the function with the profile dictionary

END
```

11. Suggested Solving Approach: Functional Approach

Use two parts:

Part 1: Profile data

Create a dictionary containing the profile information.

Part 2: Printing function

Create a function that:

- Accepts the dictionary as a parameter.
- Checks whether it is empty.
- Loops through `.items()`.
- Prints every key and value.

Suggested function structure:

```
def print_profile(profile):
    # Check for an empty dictionary
    # Loop through profile.items()
    # Print each key and value
```

This separates the **data** from the **printing logic**.

12. Easy Edge Case

Empty Dictionary

Example:

```
profile = {}
```

Because the dictionary has no items, the loop will not run.

Your function should display a helpful message:

```
Profile is empty
```

A dictionary can be checked directly:

```
if not profile:
```

This condition becomes true when the dictionary is empty.

13. Expected Input

No keyboard input is required for this problem.

The profile can be created directly in the program.

Example data:

```
name = Rahul  
phone = 9876543210  
city = Bangalore
```

Conceptually, the dictionary will contain:

```
{  
    name: Rahul,  
    phone: 9876543210,  
    city: Bangalore  
}
```

14. Expected Output

```
name: Rahul  
phone: 9876543210  
city: Bangalore
```

For an empty dictionary:

```
Profile is empty
```

15. Hint Only

Inside your function, first check whether the dictionary is empty.

Then use this loop pattern:

```
for key, value in profile.items():
```

Use one formatted `print()` statement inside the loop to place a colon between the key and value.